SSNAC: DANGs and SWINGs

Social and spatial network analysis with causal interpretation

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CTRL+F Consulting Group

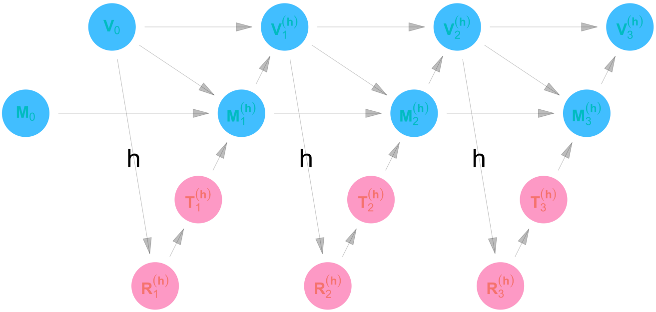
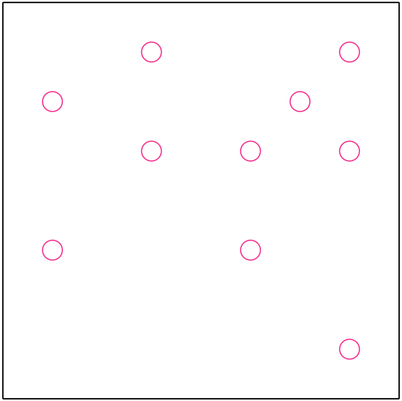


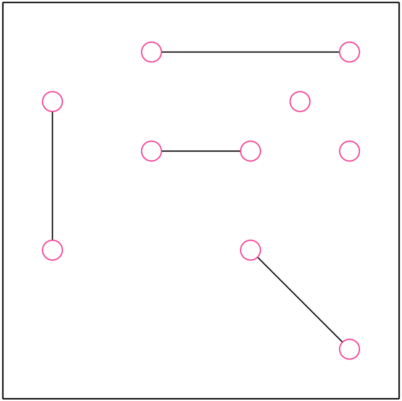
Figure 1: Dynamic regime SWING for vaccination campaign

SSNAC is a way of conceptualizing study data

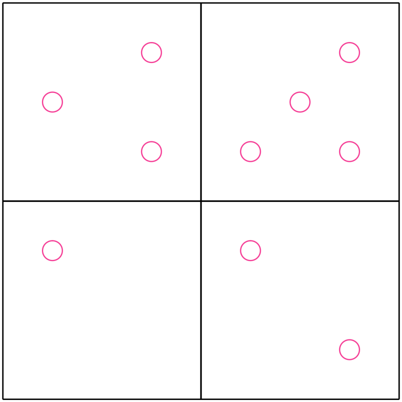
Social and spatial context



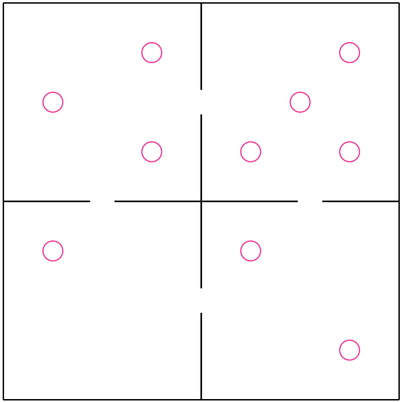
Social and spatial context



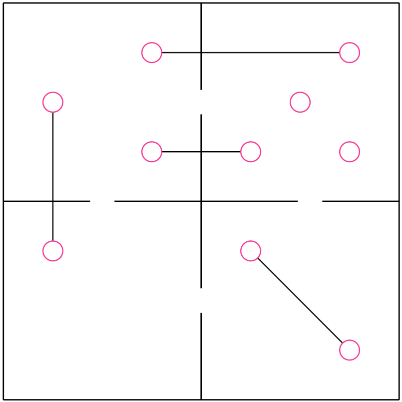
Social and spatial context



Social and spatial context



Social and spatial context



From data tables to data graphs

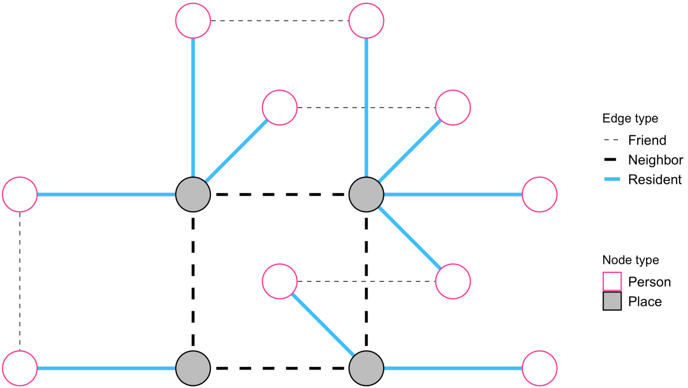


Figure 2: SSNAC data graph encoding people, places, and spatial and social connections

SSNAC is a way to reason causally in a system characterized by complex patterns of interdependence among observations

Example

We are investigating the relationship between dressing in formal wear, wealth, and job offers.

We recruit three people, *Person 1*, *Person 2*, and *Person 3*. *Person 1* and *Person 2* are married to each other. *Person 1* and *Person 3* are newly acquainted colleagues.

Non-parametric structural equation model
with independent errors

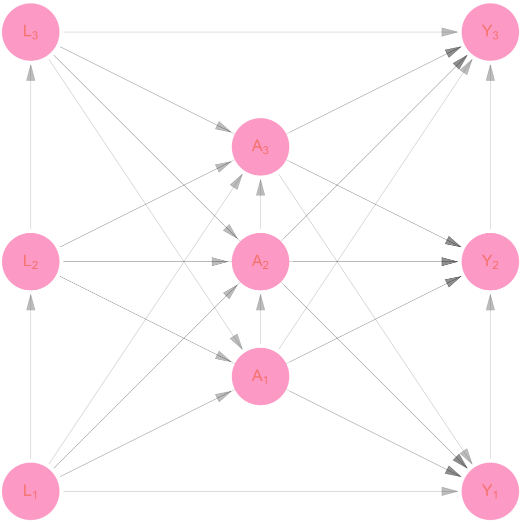


Figure 3: Fully connected causal directed acyclic graph (DAG)

NPSEM-IE

Structural equations

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}, L_1) \\
L_3 &= f_{L_3}(U_{L_3}, L_1, L_2) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, L_2, L_3) \\
A_2 &= f_{A_2}(U_{A_2}, L_1, L_2, L_3, A_1) \\
A_3 &= f_{A_3}(U_{A_3}, L_1, L_2, L_3, A_1, A_2) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, L_2, L_3, A_1, A_2, A_3) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_1, L_2, L_3, A_1, A_2, A_3, Y_1) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_1, L_2, L_3, A_1, A_2, A_3, Y_1, Y_2)
\end{aligned}
$$

Model assumptions under classic causal inference

Model assumptions under classic causal inference

Independence I: No spillover

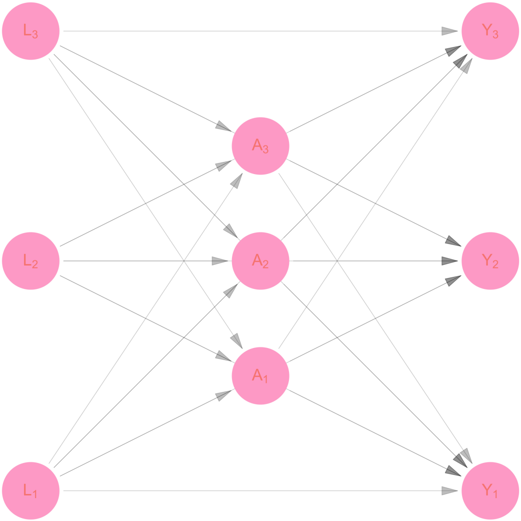


Figure 4: Causal DAG with no spillover

Model assumptions under classic causal inference

Independence I: No spillover

$$\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, L_2, L_3) \\
A_2 &= f_{A_2}(U_{A_2}, L_1, L_2, L_3) \\
A_3 &= f_{A_3}(U_{A_3}, L_1, L_2, L_3) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, L_2, L_3, A_1, A_2, A_3) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_1, L_2, L_3, A_1, A_2, A_3) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_1, L_2, L_3, A_1, A_2, A_3)
\end{aligned} \tag{1}$$

Model assumptions under classic causal inference
Independence II: No interference

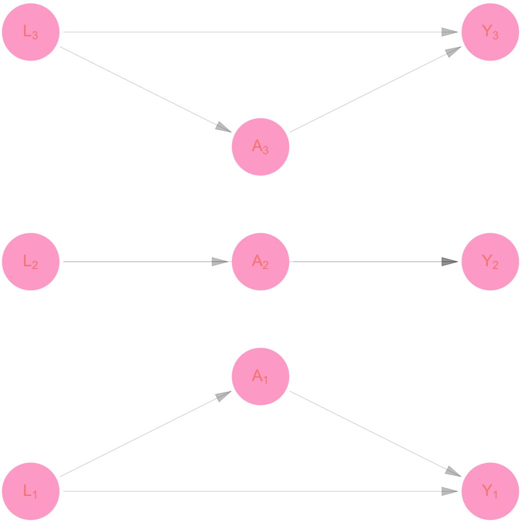


Figure 5: Causal DAG with no interference or spillover

Model assumptions under classic causal inference

Independence II: No interference

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1) \\
A_2 &= f_{A_2}(U_{A_2}, L_2) \\
A_3 &= f_{A_3}(U_{A_3}, L_3) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, A_1) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_2, A_2) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_3, A_3)
\end{aligned}
$$

Model assumptions under classic causal inference
Homogeneity I: Identical distributions

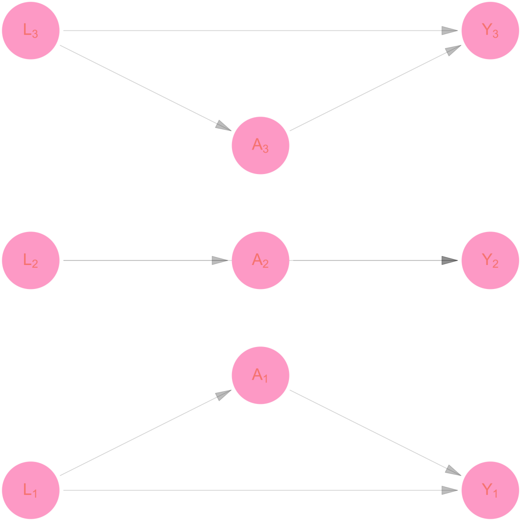


Figure 6: Causal DAG with no interference or spillover

Model assumptions under classic causal inference

Homogeneity I: Identical distributions

$$\begin{aligned} L_i &= f_L(U_{L_i}) \\ A_i &= f_A(U_{A_i}, L_i) \\ Y_i &= f_Y(U_{Y_i}, L_i, A_i) \end{aligned} \tag{2}$$

Model assumptions under SSNAC

Model assumptions under SSNAC

Citations:

- Aronow, Peter M, and Cyrus Samii. (2017). “Estimating Average Causal Effects Under General Interference, with Application to a Social Network Experiment.”
- Makofane, K., Tchetgen Tchetgen, E. J., Bassett, M. T., Berkman, L. F. (Accepted 2023. in press?). Networked wealth and mortality in the Agincourt Health and Demographic Surveillance System 2009 – 2018. American Journal of Epidemiology

Model assumptions under SSNAC
Independence I: No spillover

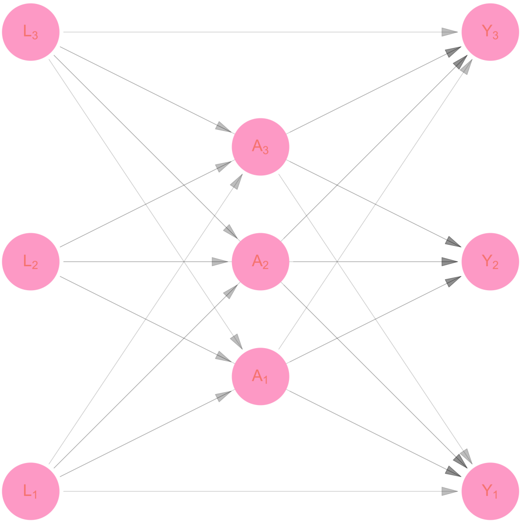


Figure 7: Causal DAG with no spillover

Model assumptions under SSNAC
Independence II: Networked interference

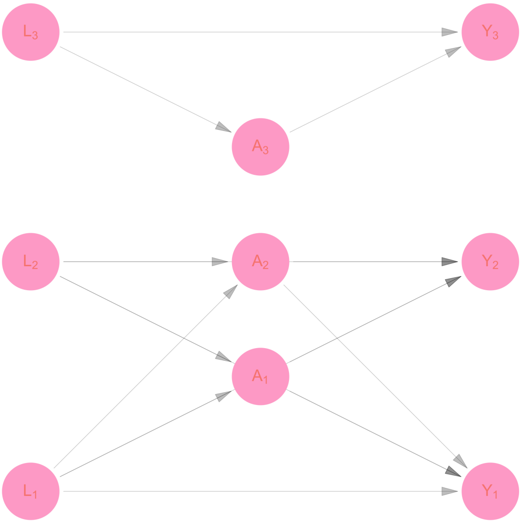


Figure 8: Causal DAG with network interference

Model assumptions under SSNAC

Independence II: Networked interference

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, L_2) \\
A_2 &= f_{A_2}(U_{A_2}, L_1, L_2) \\
A_3 &= f_{A_3}(U_{A_3}, L_3) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, L_2, A_1, A_2) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_1, L_2, A_1, A_2) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_3, A_3)
\end{aligned} \tag{3}
$$

Model assumptions under SSNAC

Homogeneity I: Separability

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, g_{A_1}(\mathbf{L}, 1)) \\
A_2 &= f_{A_2}(U_{A_2}, L_2, g_{A_2}(\mathbf{L}, 2)) \\
A_3 &= f_{A_3}(U_{A_3}, L_3, g_{A_3}(\mathbf{L}, 3)) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, A_1, g_{Y_1}(\mathbf{L}, 1), h_{Y_1}(\mathbf{A}, 1)) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_2, A_2, g_{Y_2}(\mathbf{L}, 2), h_{Y_2}(\mathbf{A}, 2)) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_3, A_3, g_{Y_2}(\mathbf{L}, 3), h_{Y_2}(\mathbf{A}, 3))
\end{aligned}
$$

for symmetric and bounded g_{A_*}, g_{Y_*}, and h_{Y_*}

Model assumptions under SSNAC

Homogeneity I: Separability

a. Symmetry

We say component function g_{ti} is symmetric with respect to data graph $\mathcal{G}$ if for any $i, j \in \mathcal{N}$, $\mathbf{X}, \mathbf{X}' \in \mathbf{R}^n$ and conformable permutation matrix $\mathbf{M}$:

$$\mathbf{X}_{\mathcal{N}_i} = \mathbf{M}\mathbf{X}'_{\mathcal{N}_j} \implies g_{ti}(\mathbf{X}, i) = g_{ti}(\mathbf{X}', j)$$

b. Boundedness

We say that an exposure mapping g_i is bounded if all possible exposure values are contained by a finite set $\mathbf{D}$

Model assumptions under SSNAC
Homogeneity I: Separability

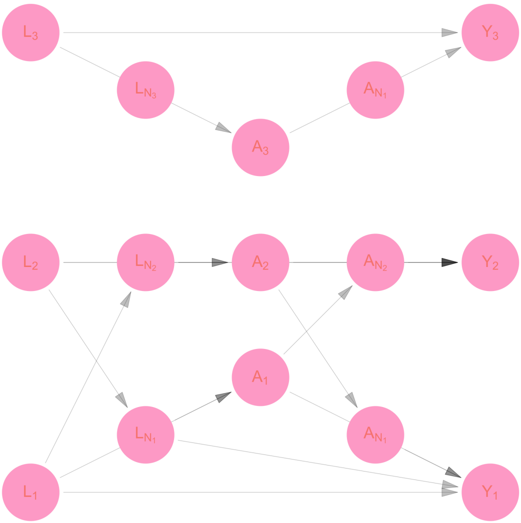


Figure 9: Causal DAG with homogenous structure

Model assumptions under SSNAC

Homogeneity II: Positional invariance

$$
\begin{aligned}
L_1 &= f_{L_1}(U_{L_1}) \\
L_2 &= f_{L_2}(U_{L_2}) \\
L_3 &= f_{L_3}(U_{L_3}) \\
A_1 &= f_{A_1}(U_{A_1}, L_1, g_A(\mathbf{L}, 1)) \\
A_2 &= f_{A_2}(U_{A_2}, L_2, g_A(\mathbf{L}, 2)) \\
A_3 &= f_{A_3}(U_{A_3}, L_3, g_A(\mathbf{L}, 3)) \\
Y_1 &= f_{Y_1}(U_{Y_1}, L_1, A_1, g_Y(\mathbf{L}, 1), h_Y(\mathbf{A}, 1)) \\
Y_2 &= f_{Y_2}(U_{Y_2}, L_2, A_2, g_Y(\mathbf{L}, 2), h_Y(\mathbf{A}, 2)) \\
Y_3 &= f_{Y_3}(U_{Y_3}, L_3, A_3, g_Y(\mathbf{L}, 3), h_Y(\mathbf{A}, 3))
\end{aligned}
$$

Model assumptions under SSNAC

Homogeneity III: Identical distributions

$$
\begin{aligned}
L_1 &= f_L(U_{L_1}) \\
L_2 &= f_L(U_{L_2}) \\
L_3 &= f_L(U_{L_3}) \\
A_1 &= f_A(U_{A_1}, L_1, g_A(\mathbf{L}, 1)) \\
A_2 &= f_A(U_{A_2}, L_2, g_A(\mathbf{L}, 2)) \\
A_3 &= f_A(U_{A_3}, L_3, g_A(\mathbf{L}, 3)) \\
Y_1 &= f_Y(U_{Y_1}, L_1, A_1, g_Y(\mathbf{L}, 1), h_Y(\mathbf{A}, 1)) \\
Y_2 &= f_Y(U_{Y_2}, L_2, A_2, g_Y(\mathbf{L}, 2), h_Y(\mathbf{A}, 2)) \\
Y_3 &= f_Y(U_{Y_3}, L_3, A_3, g_Y(\mathbf{L}, 3), h_Y(\mathbf{A}, 3))
\end{aligned}
$$

Model assumptions under SSNAC

Homogeneity III: Identical distributions

$$
\begin{aligned}
L_i &= f_L(U_{L_i}) \\
A_i &= f_A(U_{A_i}, L_i, g_A(\mathbf{L}, i)) \\
Y_i &= f_Y(U_{Y_i}, L_i, A_i, g_Y(\mathbf{L}, i), h_Y(\mathbf{A}, i))
\end{aligned}
$$

Introducing the causal Directed Acyclic Network Graph (causal DANG)

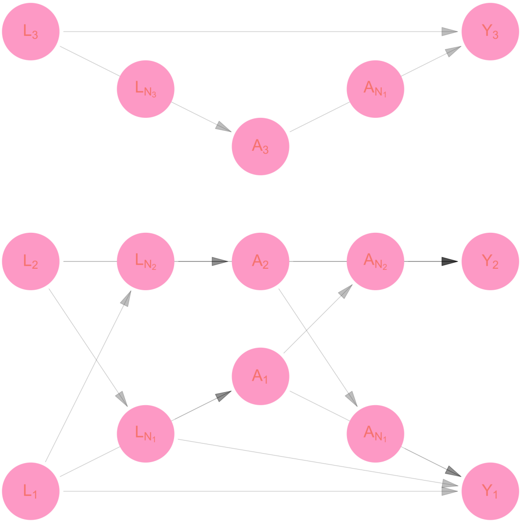


Figure 10: Causal DAG with homogenous structure

Causal Directed Acyclic Network Graph (DANG)

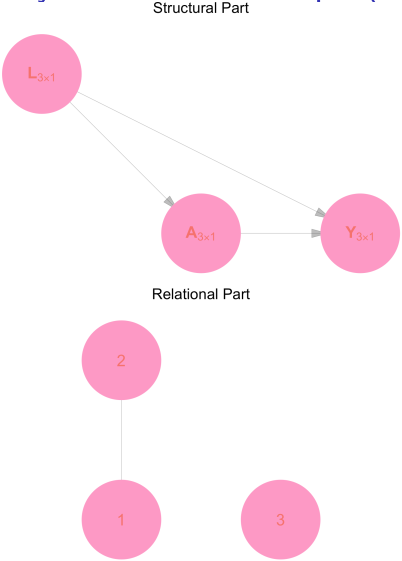


Figure 11: Causal DANG

Integrating space - data-driven vaccination campaign

Integrating space
Directed Acyclic Graph (DANG)

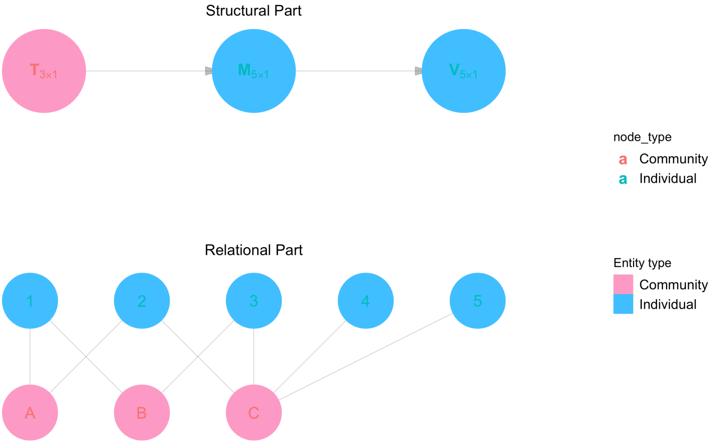


Figure 12: Causal DANG for neighborhood-level vaccination campaign

Introducing the Single World Intervention Network Graph (SWING)

Introducing the SWIG

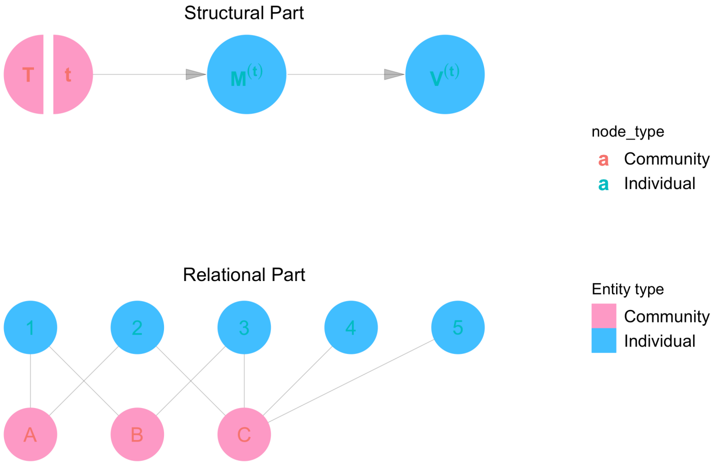


Figure 13: SWING for neighborhood-level vaccination campaign

Responding to an outbreak of infectious disease: a dynamic-regime SSNAC analysis of MPX NYC data

MPX NYC

Novel anonymous data collection tool

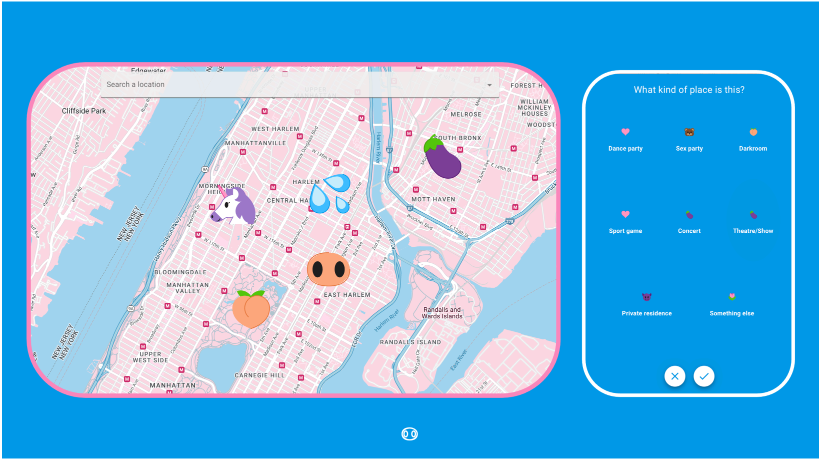


Figure 14: Map input question format developed for MPX NYC

MPX NYC

Novel tools for analyzing interaction in place

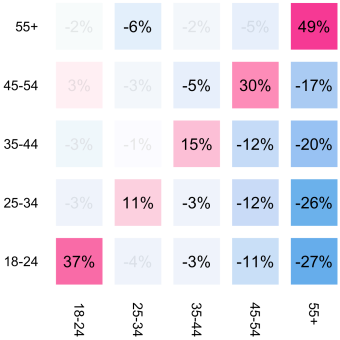


Figure 15: Spatial mixing coefficient by age group

MPX NYC

Novel dataset for social and spatial analysis

MPX NYC

Model outbreak response: contact- or movement-neutralizing?

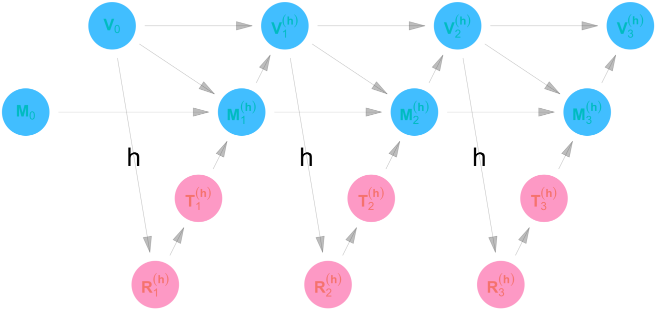


Figure 16: Dynamic regime SWING for vaccination campaign

MPX NYC

Model outbreak response: contact- or movement-neutralizing?

$$
\begin{aligned}
M_{0i} &= 0 \\
V_{0i} &= 0 \\
R_{sj}^{(h)} &= h(\mathbf{V}_{s-1}) \\
T_{sj}^{(h)} &= \begin{cases} 1 & R_{sj}^{(h)} = max_k R_{sk}^{(h)} \\ 0 & \text{otherwise} \end{cases} \\
M_{si}^{(h)} &= \begin{cases} 1 & \mathbf{T}_{s\mathcal{N}_i}^{(h)} \neq \mathbf{0} \\ 0 & \text{otherwise} \end{cases} \\
V_{si}^{(h)} &= \begin{cases} 1 & M_{si}^{(h)} = 1 \text{ or } V_{s-1,i}^{(h)} = 1 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
\tag{4}
$$

MPX NYC

Model outbreak response: contact- or movement-neutralizing?

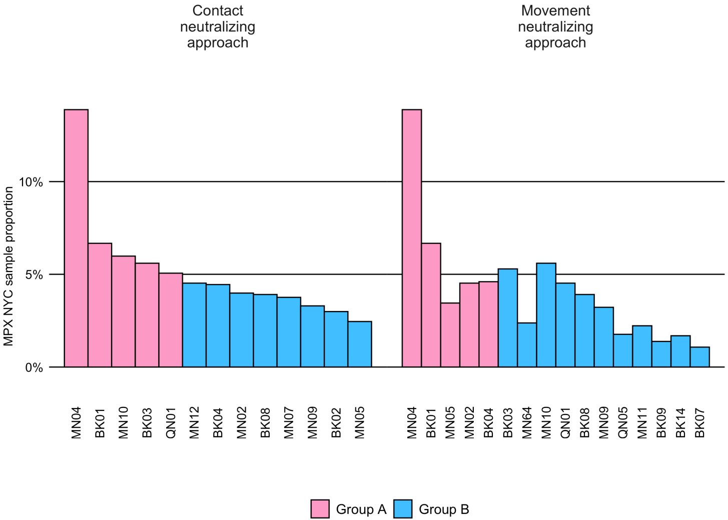


Figure 17: Priority community districts vs. coverage

MPX NYC

Model outbreak response: contact- or movement-neutralizing?

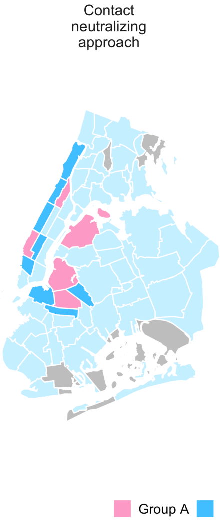


Figure 18: Priority community districts

MPX NYC main report coming soon!

Preview draft appendices for MPX NYC report

url: https://mpxnycreport.netlify.app
passw: the_final_report

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